

Development and Validation of a Questionnaire for Assessment of Happiness: The Happiness Rating Scale

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Abstract

Background. Cohort studies indicate that chronic anxiety, aggression and depression may be associated with neuroendocrine dysfunctions and lack of happiness which may predispose to poor social and physical health, leading to cardio-metabolic diseases (CMDs). Dopamine, serotonin, oxytocin and endorphins are the quartet neurotransmitters responsible for the happiness of the mind, which may be blunted by stress hormones released due to psychological diseases. This study aims to develop and validate a questionnaire for assessment of happiness to find out the prevalence of happiness in relation to coronary artery disease (CAD).

Subjects and Methods. After written informed consent and approval from the hospital ethics committee, this cross-sectional survey was conducted at Halberg Hospital and Research Institute, Moradabad, India. All subjects, 980 urban (495 men and 485 women), 900 rural (510 men and 390 women) above 25 years of age were randomly selected and recruited from urban and rural populations. Clinical data as well as risk factors were recorded with the help of case record form and validated questionnaires. Assessment of happiness was made by a new questionnaire involving happiness of mind by including questions related to social, psychological and spiritual behaviors.

Results. The prevalence of happiness was 62.4% (n = 612) among urban and 63.2% (n = 569) among rural subjects, with overall prevalence of 62.8% (n = 1,181). The prevalence of happiness was significantly greater among subjects > 45 years among both urban and rural populations compared to subjects between 25 to 45 years and the trend was significant. The overall prevalence of happiness was also significantly greater among subjects above 45 years compared to subject below 45 years of age. The prevalence of happiness overall showed a graded increase with age from 25 to 84 years in both sexes among urban and rural populations and the trends were significant (P < 0.02). The overall prevalence of tobacco intake was 16.2%, (n = 304) and alcoholism 1.96%. The frequency of tobacco intake and alcoholism were significantly lower among subjects with happiness compared to those with no happiness. However, among subjects with happiness, the frequency of moderate alcohol intake was significantly greater compared to urban subjects with no happiness. The prevalence of CAD was significantly lower among subjects with happiness

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